A Needle in Every Direction

A reader's guide to OpenAI's claimed proof of the Kakeya set conjecture, with a check of its Lean statement

Zinan Xing

Exposition only — all mathematics is due to the cited authors.

November 11, 2026

Abstract

A set that contains a unit line segment in every direction can have zero volume. The Kakeya set conjecture says that it must nevertheless have full dimension. In October 2026 OpenAI released a preprint, together with a Lean formalization, that claims this in every dimension, as a consequence of an estimate for waves. This guide is written for a reader who knows calculus and a little linear algebra, not harmonic analysis. It explains how a question about needles becomes a question about waves, what the new wave estimate says and how its proof is organized, and how the estimate comes back to needles. Separately, it compares the formal Lean statement, clause by clause, with the official problem text. Citations [P . . .] refer to the preprint [P] by page; [background: . . .] notes give context from the wider literature.

Where things stand

- **Claimed** [P Cor. 9.1, p. 27]: for every $n \geq 2$, every subset of $\mathbb{R}^n$ that contains a unit line segment in each direction has Hausdorff dimension n. For compact sets, $n = 2$ is classical and $n = 3$ is the 2025 theorem of Wang and Zahl. For $n = 4$ an earlier OpenAI preprint claimed the result by another method [P p. 27]; $n \geq 5$ is claimed here for the first time.
- **Review status.** The source is a preprint, released by OpenAI as the output of an AI model. It has not been refereed. A Lean formalization is public in the same repository; it has not been verified by MathDB, and it was not rebuilt for this guide. Section 8 says what it states.
- **Still open:** the exact strength of the estimates (a power of a logarithm is not pinned down), and the endpoint cases of the neighbouring Fourier restriction problem.

1 The question

The problem goes back to 1917 [Off]: how little area is needed to turn a needle of length one completely around in the plane? A disk of diameter one works. The surprise, due to Besicovitch, is that there is no lower limit: the needle can be turned in a region of area as small as one likes, and there are sets of area *zero* that still contain a unit segment in every direction [background: [KT], p. 1].

The official problem text keeps only the static part of the question [Off]:

(1) a *Kakeya set* is a compact set $E \subseteq \mathbb{R}^n$;

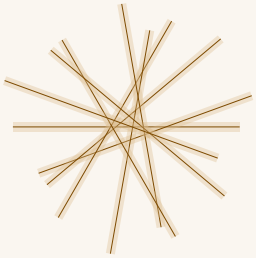
(2) for every direction e on the unit sphere S^{n-1} there is a point a such that the segment $\{a + te : 0 \le t \le 1\}$ lies in E;

(3) **conjecture:** for every $n \ge 4$, the Hausdorff dimension of E is n.

The same statement for $n = 2$ and $n = 3$ is listed there as a theorem. In the plane it is due to Davies (1971) [background: [KT], p. 3]. In space it is due to Wang and Zahl (2025), with a shorter proof by Guth, Wang and Zahl [background: [WZ], Thm. 1.1, p. 3; [GWZ], p. 1].

So the question is: a Kakeya set may have no volume at all, but is it still "n-dimensional"? The needles can be made to overlap a great deal. The conjecture says they cannot overlap so much that the set becomes as thin as a surface or a curve.

Picture, not proof



each needle is thickened to a tube of length 1 and radius δ; how small can the union be?

Fatten every needle to a thin tube of radius δ. There are about $\delta^{-(n-1)}$ essentially different directions, and each tube has volume about δ^{n-1}, so the tubes carry total volume about 1. If they barely overlapped, their union would have volume about 1. Besicovitch's sets show that the union can have volume tending to zero as $\delta \to 0$. The conjecture implies that it tends to zero *slowly*: more slowly than any power of δ.

2 Two ways to measure a thin set

Dimension is a way to compare sets of zero volume. Both definitions below are standard [background: [KT], pp. 1–2].

Counting balls of one size (Minkowski dimension). Let F_δ be the set of points within distance δ of a bounded set F. A curve of length one has $|F_\delta| \approx \delta^{n-1}$; a piece of surface has $|F_\delta| \approx \delta^{n-2}$. In general, F has Minkowski dimension d when $|F_\delta|$ behaves like δ^{n-d}. Equivalently, about δ^{-d} balls of radius δ are needed to cover F.

Balls of different sizes (Hausdorff dimension). Now allow a cover of F by balls B_j of any radii r_j, and charge each ball the price r_j^s. If, for some s, the total price $\sum_j r_j^s$ can be made as small as we like, then F is too thin to be s-dimensional. The Hausdorff dimension $\dim_{\mathrm{H}} F$ is the largest s for which this is impossible:

$$\dim_{\mathrm{H}} F = \sup\Big\{ s : \text{ there is } c > 0 \text{ with } \sum_j r_j^s \geq c \text{ for every cover of } F \text{ by balls} \Big\}.$$

Equal balls are a special case, so the Hausdorff dimension is at most the Minkowski dimension. A proof that $\dim_{\mathrm{H}} F = n$ is therefore the stronger statement, and it has to defeat covers that mix large and small balls.

A logarithm is not a power

Suppose all we know is $|F_\delta| \geq c\,[\log(1/\delta)]^{-k}$ for small δ, with fixed $c, k > 0$. If F had Minkowski dimension $d < n$, then $|F_\delta|$ would behave like δ^{n-d}. But $\delta^{n-d}[\log(1/\delta)]^k \to 0$ as $\delta \to 0$: any power of δ beats any power of the logarithm. So $d = n$. The whole proof below ends in a bound of exactly this logarithmic kind.

3 The theorem, in words and in symbols

Theorem ([P Cor. 9.1, p. 27]). *Let $n \geq 2$. Every subset of $\mathbb{R}^n$ that contains a unit line segment in each direction has Hausdorff dimension n.*

In words: however cleverly the needles are arranged, the set they sweep out is as large, in the sense of dimension, as the whole space. No compactness and no measurability of the set is assumed [P Cor. 9.1, p. 27], so the statement covers the official one and more.

The preprint obtains the theorem from a stronger, quantitative statement. For a function f on $\mathbb{R}^n$, a direction ω and $0 < \delta \leq 1$, let $K_\delta f(\omega)$ be the largest average of $|f|$ over a tube of length 1 and radius δ pointing in the direction ω. This is the *Kakeya maximal function*. The preprint states

$$\Big(\int_{S^{n-1}} K_\delta f(\omega)^n \, d\sigma(\omega) \Big)^{1/n} \;\leq\; C_n \, \big[\log(2/\delta)\big]^{2(n-1)/n} \Big(\int_{\mathbb{R}^n} |f|^n \Big)^{1/n} \qquad \text{[P Cor. 9.1, p. 27].} \tag{1}$$

The conjectured form of this inequality allows a loss $\delta^{-\varepsilon}$ for every $\varepsilon > 0$ [P p. 3]; a power of a logarithm is smaller than that. Applied to the indicator function of the neighbourhood F_δ of a Kakeya set F, it gives

$$|F_\delta| \;\geq\; c_n \, \big[\log(2/\delta)\big]^{-2(n-1)} \qquad \text{[P (9.6), p. 30].} \tag{2}$$

Indeed, F_δ contains a whole tube in every direction, so $K_\delta f = 1$ everywhere on the sphere when f is the indicator of F_δ, while the right side of (1) is a logarithm times $|F_\delta|^{1/n}$.

What the theorem does *not* say. It does not say that a Kakeya set has positive volume: Besicovitch's sets have volume zero, and (2) is consistent with that. It says nothing about the analogous problem over finite fields, which is a separate, earlier theorem [Off]. And it is not yet an accepted theorem: see the box on the first page.

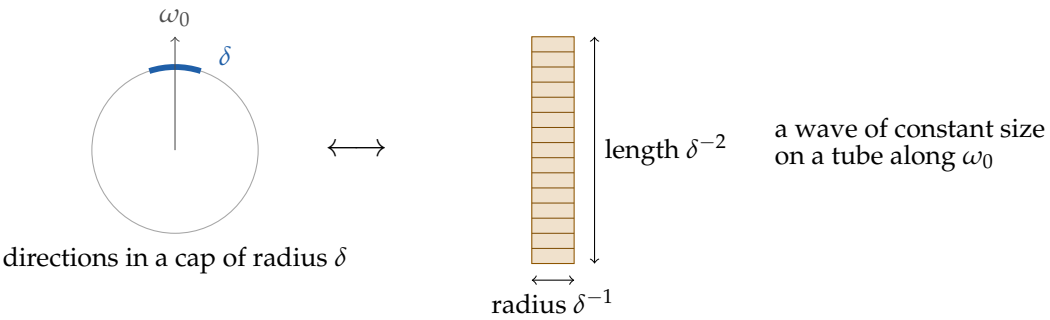
4 The key idea: trade needles for waves

The proof never works with the needles directly. It uses a dictionary, built up since the 1970s [background: [Off], Background], between thin tubes and waves.

Take a plane wave $e^{iX\cdot\omega}$: it oscillates in the direction ω and is constant on the planes perpendicular to ω. Now add up the plane waves whose directions lie in a small cap C of the unit sphere, of angular radius δ around ω_0:

$$E\mathbf{1}_C(X) = \int_C e^{iX\cdot\omega}\, d\sigma(\omega).$$

Near the origin all these waves agree, so they add up without cancellation and the sum has size about the area of the cap, δ^{n-1}. Moving away from the origin, the waves stay in step as long as their phases $X\cdot\omega$ differ by less than a fixed fraction of a turn. Along the axis ω_0 that holds out to distance about δ^{-2}; across it, out to distance about δ^{-1}.



So *a cap of directions of radius δ produces a wave of roughly constant size on a tube of length δ^{-2} and radius δ^{-1}* [P §9.2, p. 29]. Shrinking the picture by the factor δ^2 turns it into a tube of length 1 and radius δ: a fattened needle. Multiplying the cap by a suitable phase slides the tube anywhere we like [P proof of Lemma 9.4, p. 29].

This gives the plan. A family of tubes pointing in different directions corresponds to a sum of waves coming from disjoint caps. Where many tubes overlap, many waves are large at once. An upper bound for the size of such sums of waves is therefore an upper bound for how much tubes can overlap. The preprint supplies such a bound, for waves that solve the wave equation.

5 The central object: a wave that comes to a focus

Solutions of the wave equation $\partial_T^2 u = \Delta u$ in $\mathbb{R}^n$ are built from the *half-wave* $U(T)f = e^{-iT|D|}f$, which moves each plane-wave component $e^{iX\cdot\xi}$ of the initial state f at unit speed: it becomes $e^{i(X\cdot\xi - T|\xi|)}$ [P p. 1 and §1.3, p. 4]. The total $\int |U(T)f|^2$ never changes. But the wave can concentrate. A wave coming in from a thin spherical shell is spread out at first and is very large near the centre at the instant of focus [P p. 1]. Measured by $\int |u|^p$ with $p > 2$, which rewards concentration, the wave at that instant is far larger than its initial state.

The focus is brief. Sogge's *local smoothing* question asks how large the wave is when one averages over an interval of time, $1 \le T \le 2$, instead of looking at one instant [P p. 1]. The preprint's answer rests on one estimate, at the exponent

$$p_c = \frac{2n}{n-1} \qquad (p_c = 4 \text{ in the plane},\ p_c = 3 \text{ in space}).$$

Theorem (Critical estimate on one frequency band [P Thm. 1.2, p. 4]). *Let $n \geq 2$ and let f_N be made only of plane waves with wavelength about $1/N$, $N \geq 2$. Then*

$$\Big(\int_1^2 \int_{\mathbb{R}^n} |U(T) f_N|^{p_c} \, dX \, dT \Big)^{1/p_c} \;\leq\; C \, \big[\log(2+N) \big]^{2/p_c} \Big(\int_{\mathbb{R}^n} |f_N|^{p_c} \, dX \Big)^{1/p_c},$$

with a constant C that depends on n and on the band, not on N.

quantity, for wavelength $1/N$, at $p = p_c$	largest possible growth	source
size at one fixed time	a power of N: $N^{(n-1)/(2n)}$	[P p. 1]
size averaged over $1 \leq T \leq 2$	at most $[\log(2+N)]^{2/p_c}$	[P Thm. 1.2, p. 4]
size averaged over $1 \leq T \leq 2$	at least $[\log N]^{1/p_c}$, for some waves	[P Prop. 8.1, p. 26]

Averaging in time thus replaces a power of N by a power of $\log N$. The last row shows that a logarithm cannot be avoided altogether, and that the true power lies between $1/p_c$ and $2/p_c$ [P §1.3, p. 4]. Everything in Section 7 will inherit this logarithm.

6 The main steps of the wave estimate

This is the new part of the preprint [P §§2–7, pp. 6–25]. It is long and technical. We describe what each step does and where it is carried out, and leave the computations to the source.

6.1 Keep the interference: a conserved current

A focus is produced by many plane waves reinforcing one another. Any method that first takes the size of each wave and then adds the sizes loses the focus: "summing the squares would discard the interference" [P §1.4, p. 4]. The proof needs a quantity that is quadratic in the *whole* sum.

The device is to shift the wave by an imaginary amount along a light ray. A real shift only changes phases. An imaginary shift $-i\ell(1,\omega)$ multiplies the plane wave of frequency ξ by

$$\exp\big[-\tfrac{1}{2} \, \ell \, |\xi| \, \big| \omega - \xi/|\xi| \big|^2 \big] \qquad \text{[P §1.4, p. 4; §2.1, p. 6]},$$

a Gaussian that is close to 1 for waves travelling in the direction ω and tiny for the others. The shifted wave is the original one, heard through a directional filter.

From the wave continued in this way the preprint builds a matrix-valued field A on an enlarged space, in which every space-time coordinate has two auxiliary partners [P §2.3, p. 8]. Three properties matter [P Prop. 2.2, p. 9]:

- A is positive semidefinite;
- each column of A has zero divergence, which is the form of a conservation law;
- its time–time entry is a sum of two squares of the filtered wave [P (2.15), p. 10], so on a cap of directions it recovers the squared size of the wave, interference included.

The construction is done for waves on a periodic box with frequencies in one fixed cap [P §2.4, p. 10].

6.2 A potential shaped by the wave

A conserved density does not by itself control $\int |u|^{p_c}$. The proof converts density into an exponent by solving, for each value of the time variables, a nonlinear equation in the remaining variables:

$$\det H(\lambda|\Phi|^2 + u) = \lambda\mu, \qquad u = 0 \text{ on the boundary} \qquad \text{[P §1.4, p. 5]}. \tag{3}$$

Here μ is the time component of the current (the local intensity of the wave), H is an $n \times n$ matrix of second derivatives adapted to the triples of coordinates (a quaternionic Hermitian Hessian, in the preprint's terms), and $\lambda|\Phi|^2$ is a fixed background function. That this equation has a unique smooth solution is proved in an appendix [P Thm. A.1, p. 38].

Picture, not proof

Think of u as a membrane fixed at the rim and pushed down by the wave's intensity. Where the wave is concentrated, the membrane is deep. An equation of determinant type makes the depth respond to the density through a *power*, and that power is where the exponent p_c comes from.

Why the exponent is $2n/(n-1)$ [P §1.4, p. 5]

The background has curvature in only one direction: its matrix H has one eigenvalue β and $n-1$ zeros. Add a bowl of curvature α in every direction. The eigenvalues become $\beta+\alpha$ and $\alpha, \dots, \alpha$, so the determinant is $(\beta+\alpha)\,\alpha^{n-1}$. To match a density m with β fixed, one needs α of size $m^{1/(n-1)}$ (as long as α stays below β, the range the construction uses). The depth of the bowl is proportional to α. Depth times density is then $m^{1/(n-1)} \cdot m = m^{n/(n-1)}$. The density is the square of the wave, so this is the wave to the power $2n/(n-1) = p_c$.

6.3 An energy that can hardly grow

Integrating depth against density, $\int (-u)\,\mu$, and adding a correction term, gives a number $\mathcal{F}(T,\tau)$ for each time T and each value $\tau \in \mathbb{R}^2$ of the two auxiliary partners of time [P §1.4, p. 5; Lemma 4.3, p. 17]. Conservation of the current yields a balance law for $\mathcal{F}$: its Laplacian in the three variables (T,τ) can be positive only through wave flux across the side wall of the region [P Prop. 4.2, p. 16]. There the directional filter is exponentially small. With $h = 1/N$ the wavelength, a a fixed multiple of h, and $\rho = |\tau|$, the outcome is three bounds, for data normalized by $\int |f|^2 = 1$ [P Prop. 5.1, p. 18]:

$$\mathcal{F}(0,\tau) \le C\,a^2 \int |f|^{p_c} \qquad \text{on the circle } \rho = 4a \text{ at time } 0,$$

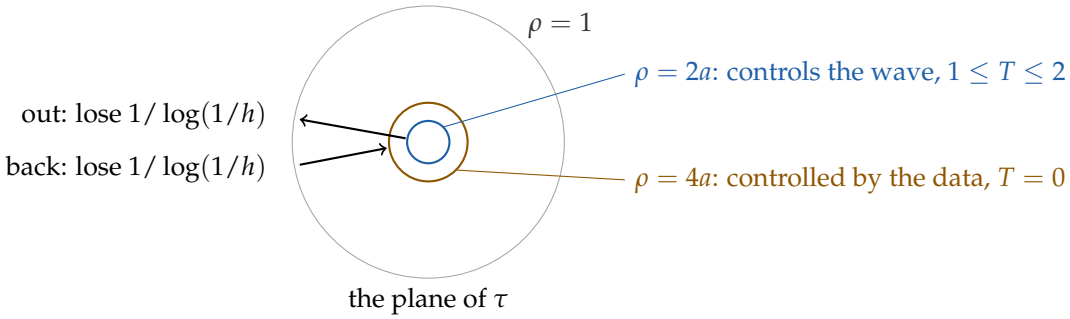
$$\mathcal{F}(T,\tau) \ge c\,a^2 \int |U(T)f|^{p_c} \qquad \text{on the circle } \rho = 2a \text{ at every time } T,$$

$$\Delta\mathcal{F} \le C\,\rho^{-2} e^{-c\rho/h}\,\mathcal{F} \qquad \text{for } 2a \le \rho \le 3.$$

The first two say that $\mathcal{F}$ measures, up to constants, the quantities in the theorem. The third says that $\mathcal{F}$ is almost *superharmonic*: its Laplacian is at most a multiple of itself that dies off exponentially away from the inner circle, fast enough not to affect the argument.

6.4 Two circles, two logarithms

A positive superharmonic function cannot be large in one place and tiny nearby. That principle carries the lower bound on the small circle $\rho = 2a$, at the observed times $1 \le T \le 2$, to the small circle $\rho = 4a$ at time 0. But the two sets are not near each other on the scale a: they are a unit of time apart. The route goes out to radius $\rho \approx 1$, across in time, and back in [P §6, p. 21].



Each radial passage costs one logarithm, for a reason from calculus: a harmonic function in the plane that depends only on the radius has the form $c_1 + c_2 \log \rho$ [P §1.4, p. 5]. Going from radius $a \approx h$ to radius 1 changes $\log \rho$ by about $\log(1/h)$. The precise statement is a lemma about any nonnegative function with $\Delta E \le VE$ [P Lemma 6.1, p. 21]. Chaining the bounds,

$$\int_1^2 \int |U(T)f|^{p_c}\, dX\, dT \;\lesssim\; a^{-2} \int_1^2 \overline{\mathcal{F}}(T, 2a)\, dT$$
$$\lesssim\; \big[\log(2/h)\big]^2 a^{-2}\, \overline{\mathcal{F}}(0, 4a) \;\lesssim\; \big[\log(2/h)\big]^2 \int |f|^{p_c}\, dX \qquad \text{[P pp. 4–5]},$$

where the bar denotes the average over a circle and $\lesssim$ hides a constant. Taking the p_c-th root gives the power $2/p_c$ of the logarithm [P Prop. 6.3, p. 23].

6.5 From the periodic box to all of space

A fixed number of caps covers all directions, so the estimate holds for a full band of frequencies on the periodic box. Waves of wavelength h travel at unit speed, and their influence far from the light cone is smaller than any power of h; this removes the periodicity and proves the critical estimate on $\mathbb{R}^n$ [P §7.1, p. 24]. The preprint then interpolates and sums over frequency bands to obtain local smoothing for all exponents $2 \le p < \infty$, with an arbitrarily small loss [P Thm. 1.1, p. 1; §7.2, pp. 24–25].

7 Finishing the proof: from waves back to needles

This part follows a classical route, which the preprint credits to earlier authors at each step [P §§9–9.3, pp. 27–31]. Write $\theta = 2/p_c$ for the power of the logarithm in the wave estimate, and $L(R) = \log(2 + R)$.

7.1 A focus sees the sphere

Send in a spherical wave whose amplitude $g(\omega)$ depends on the direction it comes from. Near the focus, magnified to the scale of one wavelength, the wave looks like the superposition of

plane waves from Section 4,

$$Eg(X) = \int_{S^{n-1}} e^{iX\cdot\omega} g(\omega)\, d\sigma(\omega) \qquad \text{[P §9, p. 27]}.$$

The wave estimate therefore gives, on every ball of radius $R \geq 2$,

$$\Big(\int_{B(z,R)} |Eg|^{p_c}\Big)^{1/p_c} \;\leq\; C\,L(R)^{\theta}\Big(\int_{S^{n-1}} |g|^{p_c}\Big)^{1/p_c} \qquad \text{[P Lemma 9.2, p. 28]}. \tag{4}$$

7.2 Random signs turn waves into tubes

Take tubes T_j of length 1 and radius about δ, whose directions ω_j are separated by at least δ, placed anywhere, and nonnegative weights w_j. By Section 4, tube T_j is where the wave Eh_j from a cap C_j around ω_j is large (after magnifying by δ^{-2}). Form

$$g = \sum_j \varepsilon_j\, w_j^{1/2}\, h_j, \qquad \varepsilon_j = \pm 1 \text{ chosen by independent coin flips.}$$

What the coin flips do

For two numbers, $(a+b)^2$ and $(a-b)^2$ average to a^2+b^2: the cross term cancels. In the same way, the average over the signs of $|Eg|^2$ is $\sum_j w_j |Eh_j|^2$ at every point, which is at least a constant times $\delta^{2(n-1)} \sum_j w_j \mathbf{1}_{T_j}$. The interference that the wave estimate had to control is averaged away, and what remains counts overlapping tubes.

Apply (4) to g for each choice of signs and average. (The preprint first groups the tubes by the unit cube containing their centres, so that each group, magnified, lies in one ball of radius about δ^{-2}.) On the right, the caps are disjoint, so $\int |g|^{p_c} = \sum_j w_j^{p_c/2} \sigma(C_j)$ whatever the signs are. On the left, the average of $|Eg|^{p_c}$ is at least the $(p_c/2)$-th power of the average of $|Eg|^2$. With $q = p_c/2 = n/(n-1)$ this gives the *overlap bound*

$$\Big(\int_{\mathbb{R}^n} \Big(\sum_j w_j \mathbf{1}_{T_j}\Big)^q\Big)^{1/q} \;\leq\; C\,L(1/\delta)^{2\theta}\Big(\delta^{n-1}\sum_j w_j^q\Big)^{1/q} \qquad \text{[P Lemma 9.4, p. 29]}. \tag{5}$$

All powers of δ cancel, and they do so only at the exponent p_c [P p. 30]. The logarithm appears with power 2θ because a tube corresponds to the *square* of a wave [P §9.2, p. 29].

The overlap bound already shows that the tubes fill space

Take one tube in each of about $\delta^{-(n-1)}$ separated directions, with all $w_j = 1$, and let U be their union. Then $\int \sum_j \mathbf{1}_{T_j} \approx 1$, and (5) says $\big(\int (\sum_j \mathbf{1}_{T_j})^q\big)^{1/q} \leq C\,L(1/\delta)^{2\theta}$. Hölder's inequality on U, with the conjugate exponent n of q, gives

$$1 \approx \int_U \sum_j \mathbf{1}_{T_j} \;\leq\; |U|^{1/n}\Big(\int \Big(\sum_j \mathbf{1}_{T_j}\Big)^q\Big)^{1/q} \;\leq\; C\,|U|^{1/n} L(1/\delta)^{2\theta}.$$

So $|U| \geq c\,L(1/\delta)^{-2\theta n} = c\,L(1/\delta)^{-2(n-1)}$, which is the shape of (2). (This shortcut is ours; the preprint argues through the maximal function, as follows.)

7.3 The maximal function and the volume bound

The maximal estimate (1) is the dual form of (5): one tests the values $K_\delta f(\omega_j)$ against weights w_j, moves the weights onto the tubes, and applies Hölder's inequality and (5) [P proof of Cor. 9.1, p. 30]. Its power of the logarithm is $2\theta = 4/p_c$, which equals $2(n-1)/n$. The volume bound (2) follows as explained after it, and by the toy computation of Section 2 it shows that a bounded Kakeya set has Minkowski dimension n [P p. 30].

7.4 Balls of different sizes

For the Hausdorff dimension one must rule out economical covers that mix scales: "we have to allow balls of different sizes in the same cover" [P p. 30]. Let F be a Kakeya set, fix $0 < s < n$, and take any cover of F by balls B_j of radii r_j. A cover that uses a ball of radius above $1/e$ already costs at least e^{-s}, so we may assume that every $r_j \le 1/e$.

Sort the balls by scale. Put ball j in group k when $2^k \le \log(1/r_j) < 2^{k+1}$, and let $\delta_k = e^{-2^{k+1}}$, which is smaller than every radius in group k. Let G_k be the union of the doubled balls of group k [P p. 30].

Look along one needle. Fix a direction ω and the unit segment of F in that direction. Each of its points lies in some ball, hence in some group. If a point lies in a ball of group k, the whole disk of radius δ_k across the segment at that point lies in the doubled ball. So the part of the segment covered by group k has length at most a constant times the average of $\mathbf{1}_{G_k}$ over the δ_k-tube around the segment, which is at most $C\,K_{\delta_k}\mathbf{1}_{G_k}(\omega)$. The lengths add up to at least 1:

$$1 \;\le\; C\sum_k K_{\delta_k}\mathbf{1}_{G_k}(\omega) \qquad \text{for every direction } \omega \qquad \text{[P pp. 30–31]}.$$

Measure over all directions. Take the L^n norm over the sphere on both sides; the norm of a sum is at most the sum of the norms. Then apply the maximal estimate (1) to each term. Since $\log(2/\delta_k) \approx 2^{k+1}$ and the volume of G_k is at most a constant times $\sum_{j\in\text{group }k} r_j^n$,

$$1 \;\le\; C\sum_k 2^{2\theta k}\Big(\sum_{j\in\text{group }k} r_j^n\Big)^{1/n}.$$

A power beats a logarithm. In group k every radius is at most e^{-2^k}, so $r_j^n \le e^{-(n-s)2^k}\, r_j^s$. The factor $2^{2\theta k}$ is a power of $\log(1/r_j)$; the factor $e^{-(n-s)2^k/n}$ is a power of r_j itself, and their products have a finite sum over k. Therefore

$$1 \;\le\; C_s\Big(\sum_j r_j^s\Big)^{1/n}.$$

The total price $\sum_j r_j^s$ of the cover cannot be small. By Section 2, $\dim_{\mathrm{H}} F \ge s$ for every $s < n$, so $\dim_{\mathrm{H}} F = n$. This is the toy computation of Section 2 again, now with all scales at once. The preprint runs the same argument with a finer price function in place of r^s and so obtains a slightly sharper conclusion [P pp. 30–31]; the grouping by scale is credited there to Keich.

8 What the Lean formalization states

8.1 How the trust is arranged

The repository contains a formal development in the Lean proof assistant, built on the Mathlib library [L]. It is organized for a checker called Comparator. A short *challenge* file states six

theorems and leaves each proof as a placeholder; it imports nothing but Mathlib. A separate *solution* module must prove the same six statements, and a configuration file allows only Lean's three standard axioms [L]. Once the checker accepts the solution, a reader who wants to know *what* was proved needs to read only the challenge file and the Mathlib definitions it uses, here Euclidean space and the Hausdorff dimension `dimH`.

The fifth of the six statements is the Kakeya set conjecture. The solution module derives it from the formal local smoothing theorem by way of the restriction and maximal estimates [L]; its supporting files run to roughly 180,000 lines.

8.2 Statement audit: the Lean text against the official text

The formal statement reads: for all $n \geq 2$ and every set F in `EuclideanSpace ℝ (Fin n)`, if for every ω with $\|\omega\| = 1$ there is an x with $\{x + t\omega : -\frac{1}{2} \leq t \leq \frac{1}{2}\} \subseteq F$, then `dimH F = n` (`LocalSmoothing.lean`, lines 54–66 and 129–131 [L]). MathDB has its own frozen Lean target for this problem [Off]; we compare all three.

No.	Official text [Off]	MathDB Lean target	OpenAI Lean statement
(1)	$E \subseteq \mathbb{R}^n$ compact	any set `S`	any set `F`
(2)	every direction $e \in S^{n-1}$	every `v` with $\|v\| = 1$	every ω with $\|\omega\| = 1$
(2)	segment $\{a + te : 0 \leq t \leq 1\}$	`affineSegment ℝ a (a + v)`	$\{x + t\omega : \|t\| \leq \frac{1}{2}\}$
(3)	$n \geq 4$	every $n > 0$	every $n \geq 2$
(3)	$\dim_{\mathrm{H}} E = n$	`dimH S = n`	`dimH F = n`

Two differences are visible, and neither has mathematical content. The segments are parametrized from an endpoint in one text and from the midpoint in the other; putting $x = a + \frac{1}{2}v$ identifies them. And the MathDB target includes $n = 1$, where the claim is elementary (a set on the line that contains an interval has dimension one), while the OpenAI statement starts at $n = 2$. Both Lean statements drop compactness, so both are stronger than the official prose, and both use the same Mathlib definition of Hausdorff dimension. Apart from the case $n = 1$, the OpenAI statement implies the MathDB target. This is a reading-level check by the author, not reviewed by a Lean expert.

8.3 What this does not certify

- **That the Lean proof checks.** OpenAI's catalogue lists it as passing its Comparator configuration [L]. We read the statement and the top of the proof; we did not rebuild it. MathDB's page for the problem records no verified formal proof [Off].
- **The logarithmic bounds.** The formal Kakeya maximal statement has a loss $\delta^{-\varepsilon}$ for every $\varepsilon > 0$ [L]. The sharper inequalities (1) and (2), with a power of a logarithm, are not among the six formal statements.
- **The prose.** A formal proof certifies a formal statement. It does not certify that each sentence of the 59-page preprint is correct as written, and the preprint itself has not been refereed. Its repository describes the manuscripts as raw research data produced by an AI model, not as polished papers [L].

9 What remains open

- **Independent confirmation.** Refereeing of the preprint, and verification of the Lean proof against MathDB's target.
- **The right logarithm.** For the wave estimate the true power lies between $1/p_c$ and $2/p_c$ [P §1.3, p. 4]. For the maximal function it lies between $1 - 1/n$ and $2(n-1)/n$ [P pp. 27 and 31].
- **Fourier restriction.** The same wave estimate gives the restriction estimates for the sphere throughout the interior of the conjectured range of exponents, not on its boundary [P p. 3; Cor. 9.3, p. 28].

(Prepared with the assistance of AI, under the direction of the author. Released under CC BY 4.0.)

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[L] OpenAI, *openai/math*, repository at commit `4eaee87`: `README.md`; `lean/ComparatorChallenges/LocalSmoothing.lean` and `.json`; `lean/OAI/MeasureTheory/Kakeya/Main.lean`; `lean/docs/079.md`; `lean/formalization.yaml`. https://github.com/openai/math/tree/4eaee87548250b2682affda7d05d947c05554bae.

[Off] MathDB, *Kakeya set conjecture in dimensions $n \geq 4$*, problem #315983, https://mathdb.com/p/315983/kakeya-conjecture; Lean target `Sun_Kakeya_kakeya_set_conjecture_proof.lean`, https://github.com/mathdb-lean/problems/tree/4c3a9ececb405c5fb50d2682a4a4cd47c544efa0/problems/315983.

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